

JKBOSE 2026

CLASS 11TH

Photosynthesis in Higher Plants

LECTURE-03

BOTANY

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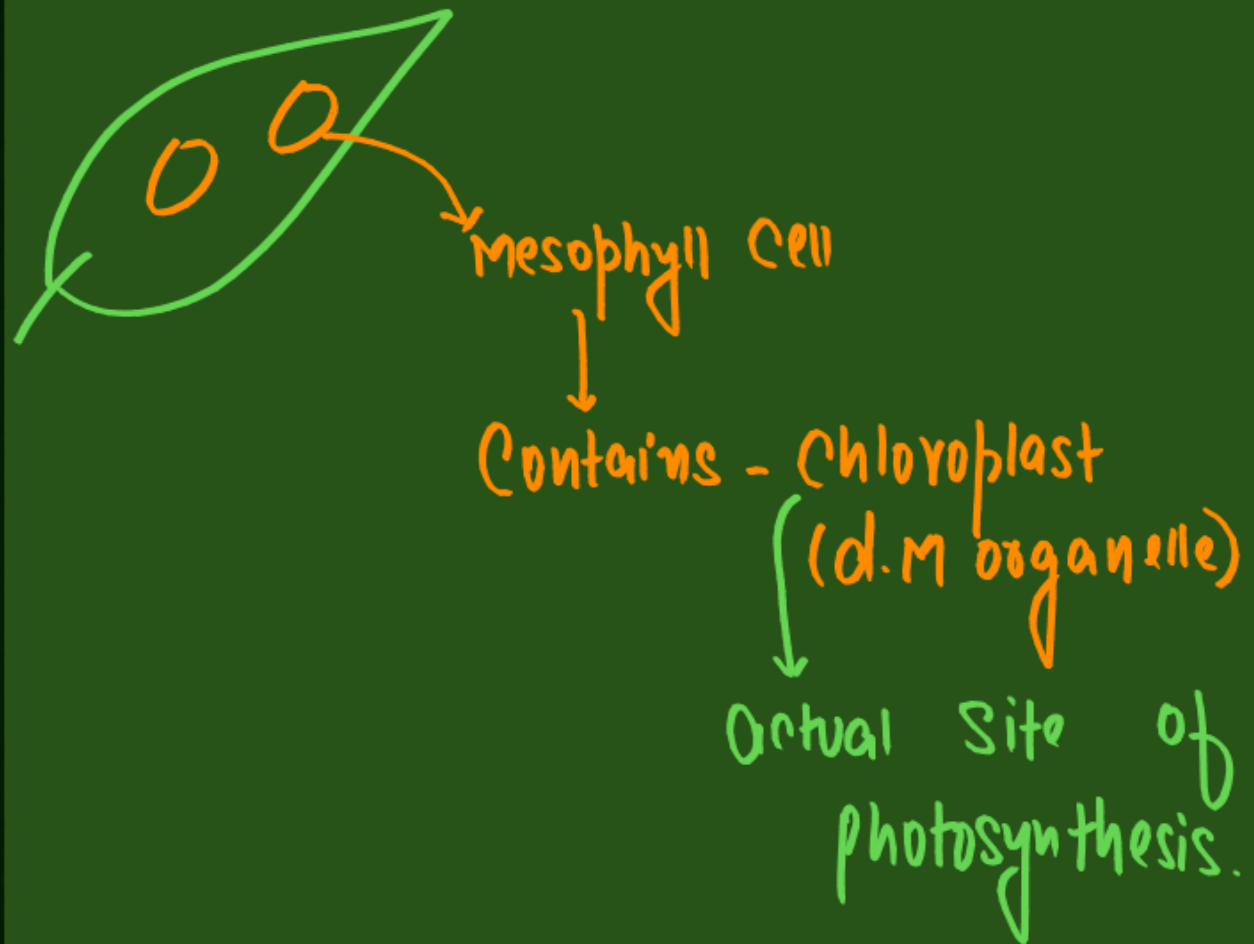


Topics To Covered

- 1** Site of photosynthesis
- 2** Pigments involved in photosynthesis

Site of photosynthesis

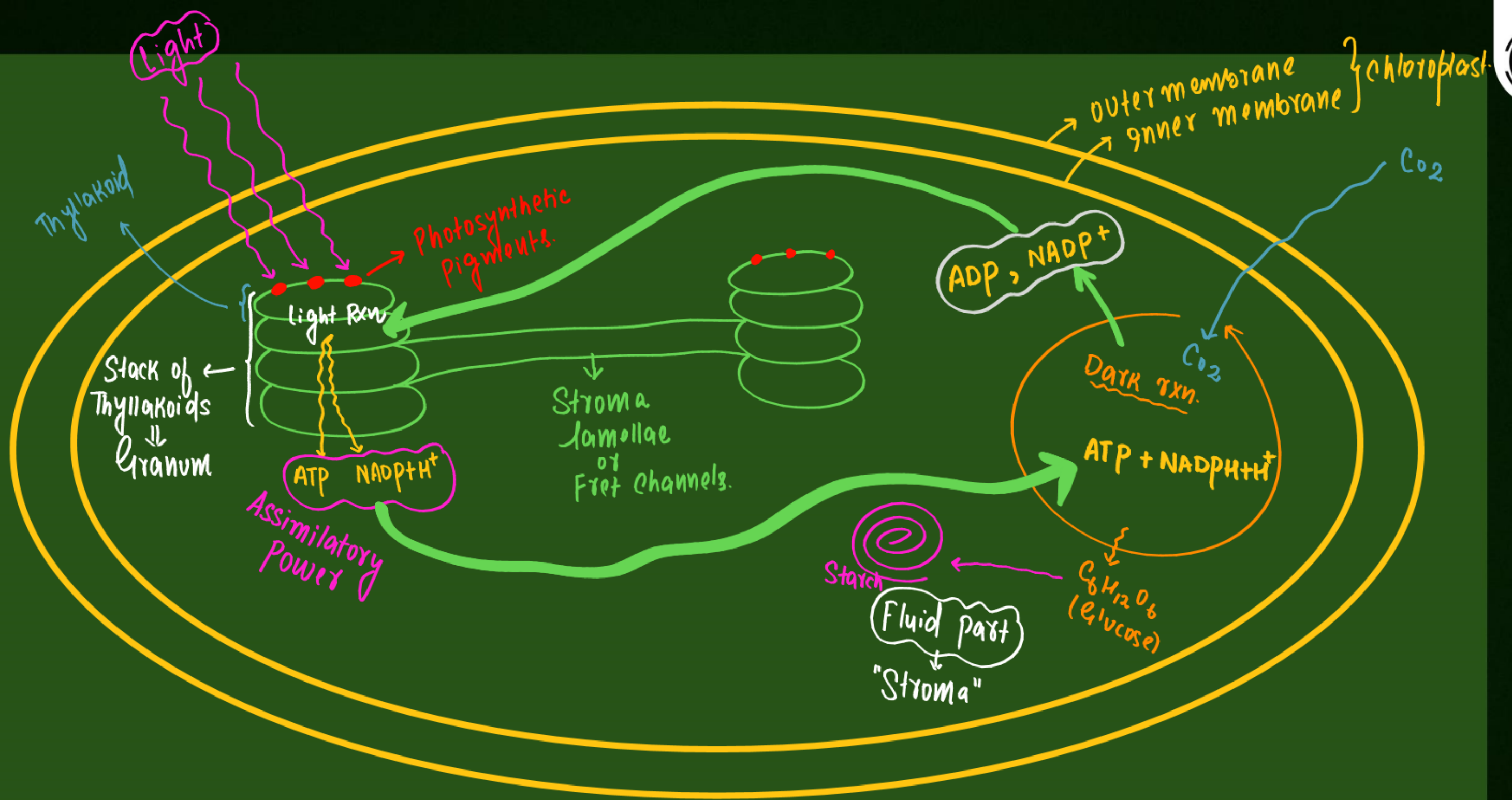
∴ In leaf, there are present **Mesophyll cells**



Inside mesophyll cells, there are present double membranous cell organelle called as

Chloroplast

This is actual site of photosynthesis.



→ Within the Chloroplast, there is division of labour.

Membranous part
(Grana thylakoid + Stroma lamellae)

- ≡ It is the site of light reaction / Photochemical reaction.
- ≡ Light rxn. directly depends on light.
- ≡ Assimilatory power (ATP & NADPH + H⁺) are formed.

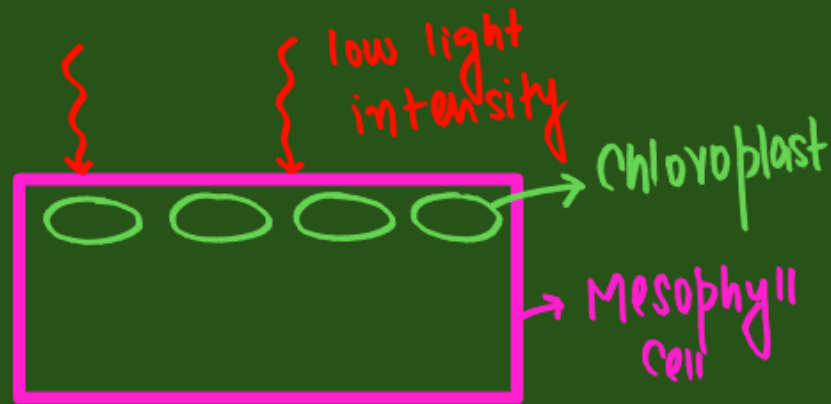
Fluid part
(Stroma)

- It is a site of Dark rxn.
- It is also called as Biosynthetic phase because glucose (C₆H₁₂O₆) is formed.
- This glucose gets stored as starch.
- Dark rxn. does not depend on light rather it depends on products of light rxn. i.e. ATP & NADPH + H⁺.

Alignment of Chloroplast Within Mesophyll

In order to obtain optimum sunlight, Chloroplast attain different alignments inside mesophyll:

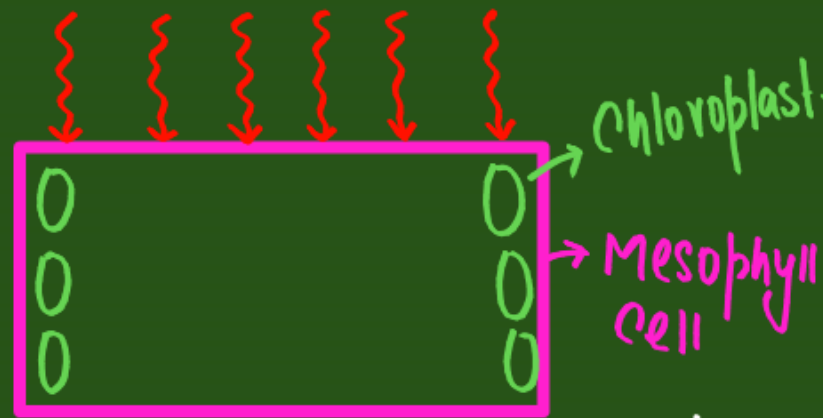
A) Under low light intensity:



Chloroplast aligns themselves perpendicular to incident sunlight (to obtain optimum sunlight).

This condition is called as **Epistrophe**.

B) Under high light intensity:

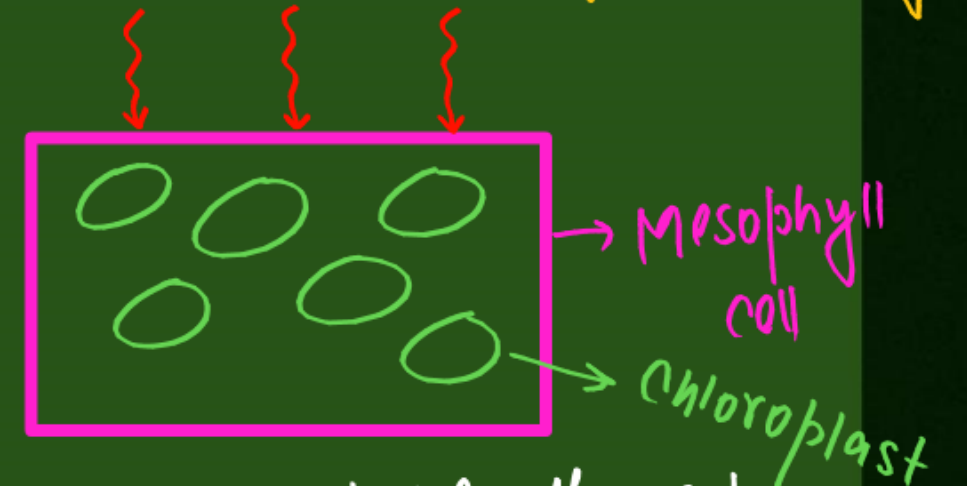


Chloroplast aligns themselves along the lateral walls of mesophyll cell.

This prevents **photo-oxidation** of chloroplast.
 (photo-oxidation leads to degradation of chloroplast)

This condition is kld as **Parastrophe**

C) Under moderate light intensity:

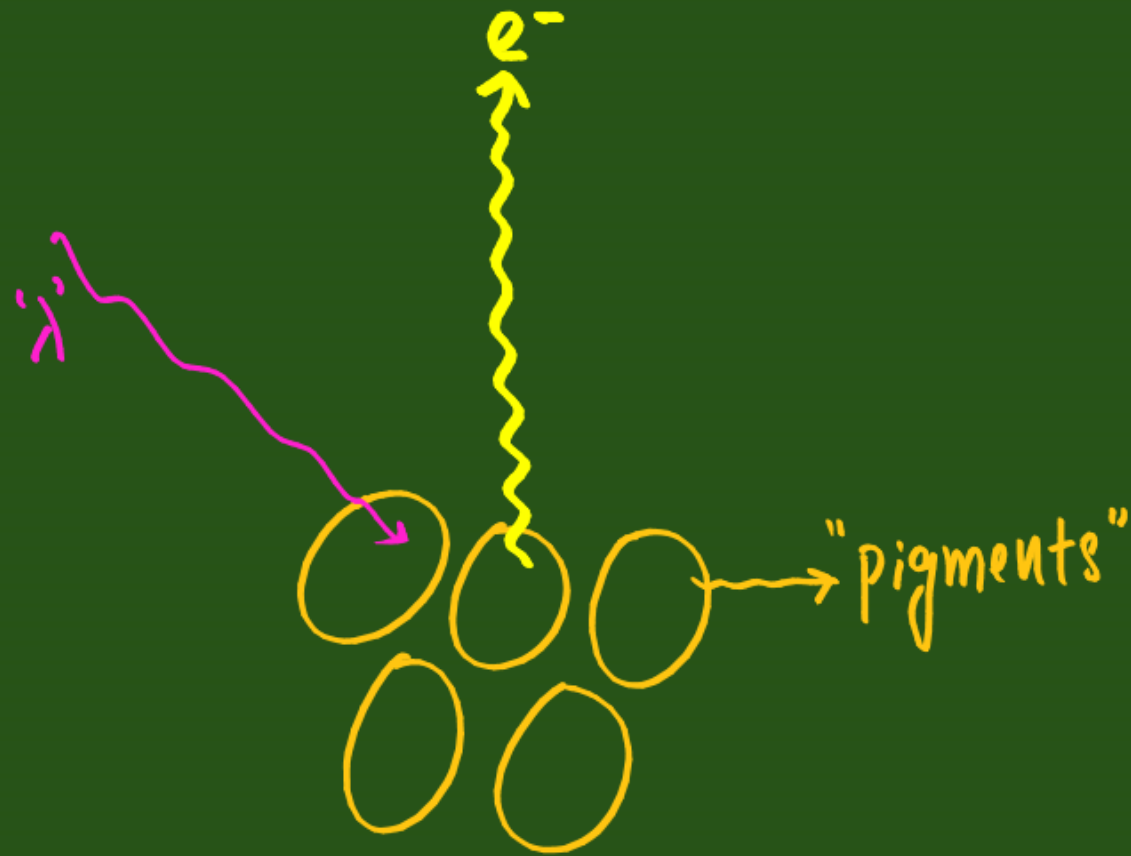


Chloroplast aligns themselves randomly inside mesophyll cells.

This condition is called as **Apostrophe**.

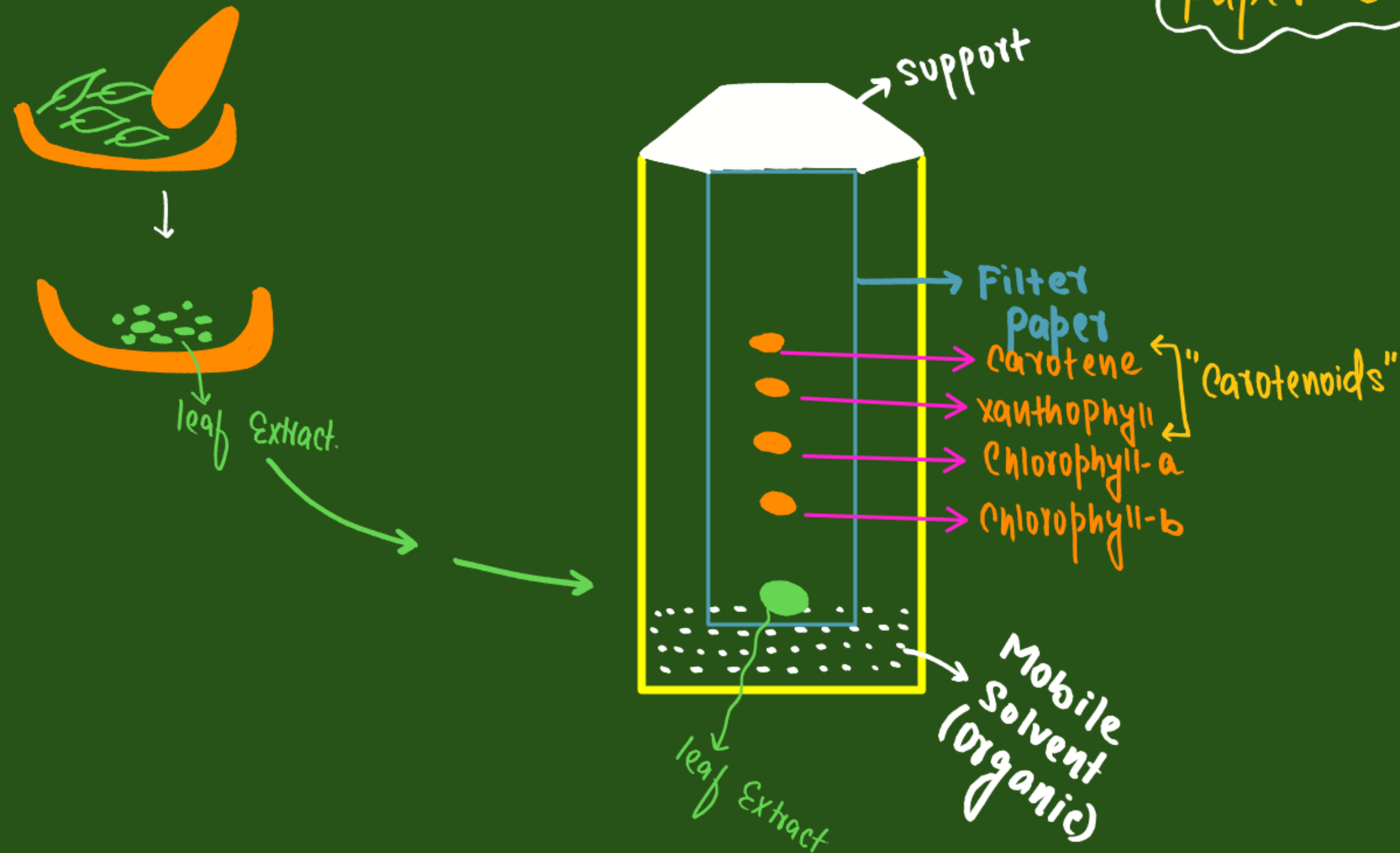
PIGMENTS INVOLVED IN PHOTOSYNTHESIS

∴ pigments are the chemical substances — which excite electrons after absorbing a light of specific wavelength.



⊗. Different shades of green in leaves of different plant or same plants is because of different types of pigments.

• These various photosynthetic pigments of leaves — can be separated by a technique called as **Paper Chromatography**



(Note) The picture obtained through paper Chromatography is called as **Chromatogram**.

∴ Thus different shades of green in leaves are due to following 04 pigments:

a. Chlorophyll-a → (Blue-green)

b. Chlorophyll-b → (Yellow-green)

c. Xanthophyll → (Yellow)

d. Carotenes → (Yellow-orange)

Carotenoids {

NOTE . Chlorophyll-a → Chief / main photosynthetic pigment.

. Chlorophyll-b
. Xanthophyll
. Carotenes } → Accessory photosynthetic pigments.

CHLOROPHYLL-a

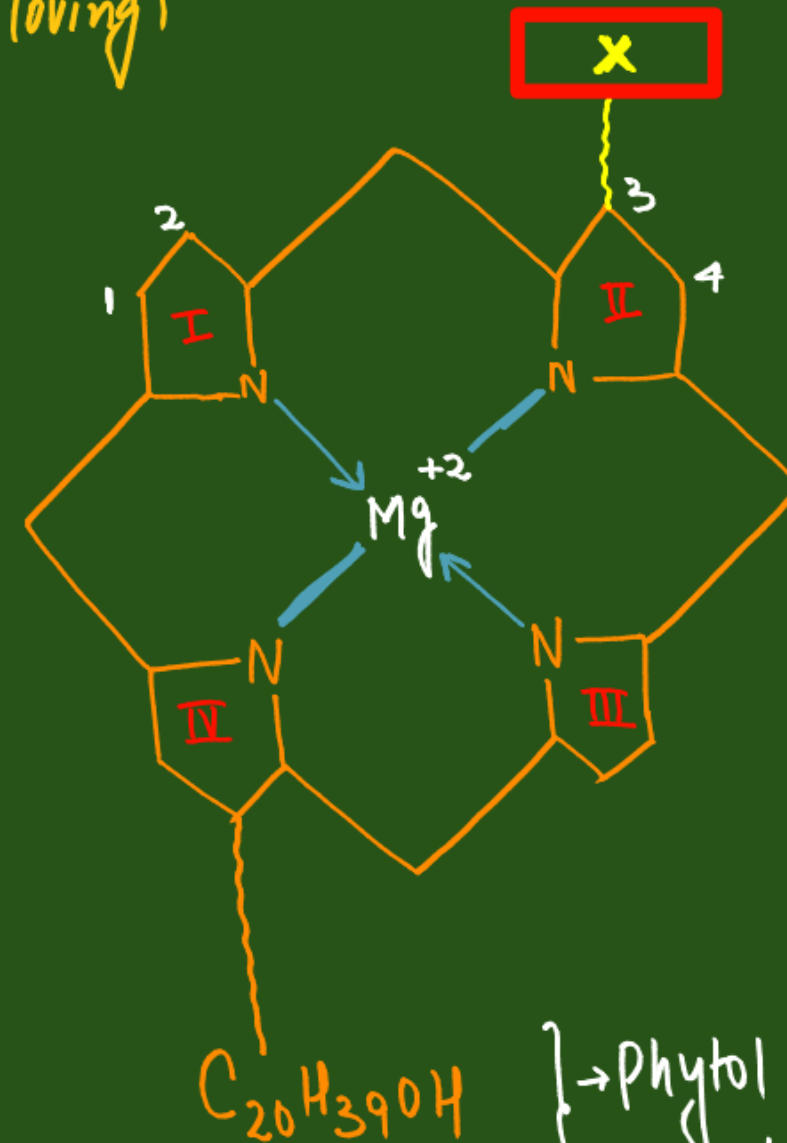
- It is chief/main photosynthetic pigment.
- It is most abundant because it is present in all the photosynthetic organisms.
- It is primary photosynthetic pigment because it carries out primary reactions like formation of ATP & $\text{NADPH} + \text{H}^+$.

• Structure of Chl-a is just like **tad-pole larva** (Assymetrical Str.)

• Two parts $\begin{cases} \text{Porphyrin Head} \rightarrow \text{Hydrophillic (H}_2\text{O loving)} \\ \text{Phytol tail} \rightarrow \text{Hydrophobic (H}_2\text{O repelling)} \end{cases}$

If $x = -\text{CH}_3 \rightarrow \text{Chlorophyll-a}$

If $x = -\text{CHO} \rightarrow \text{Chlorophyll-b}$



Porphyrin Head
Contains 04 pyrrol rings
ie, why it is called
as tetrapyrrol head.

$\left. \begin{array}{l} \rightarrow \text{Phytol tail} \\ \rightarrow \text{Alcoholic (contains -OH group)} \end{array} \right\}$

CHLOROPHYLL-b

- It is Structurally Similar to Chlorophyll-a except $\rightarrow$ at Carbon-3 of 2nd pyrrole ring, there is $-CHO$ group.
- It is accessory photosynthetic pigment.

NOTE • Centrally located Mg^{+2} — provides colour to Chlorophyll.
 • Without $Mg^{+2} \rightarrow$ Chlorophyll becomes colourless which is called as **pheophytin**.

CAROTENOIDS

- Accessory photosynthetic pigments.
- Most stable.
- 2 types :

a) Carotenes: $(C_{40}H_{56})$

Examples : α -Carotene
 β -Carotene
Lycopene — +nt in chillies, tomato.

- Most Common — β carotene
↓
+nt in carrot.

b) Xanthophylls: $(C_{40}H_{56}O_x)$ "x=1-8"

- These are oxygenated carotenes.

• Ex. Cryptoxanthin,

Fucoxanthin $\rightarrow C_{40}H_{56}O_6$
present in Brown algae.

QUESTIONS

Chlorophyll-a molecule at its carbon atom 3 of the pyrrole ring II has one of the following: (1997)

- A** Carboxylic group
- B** Magnesium
- C** Aldehyde group
- D** Methyl group

QUESTIONS

Chlorophyll a occurs in:

(1992)

- A** All photosynthetic autotrophs
- B** In all higher plants
- C** All oxygen liberating autotrophs
- D** All plants except fungi

QUESTIONS

H/W



Photosynthetic pigments found in the chloroplasts occur in:

(1991)

- A** Thylakoid membranes
- B** Plastoglobules
- C** Matrix
- D** Chloroplast envelope

HOMework

2. Question
+ Revision —.





Thank
You